A novel dataset for Vietnamese scam calls detection

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Tóm tắt nội dung

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1 Introduction

Telecommunication fraud has become a critical national issue in Vietnam, as fraud techniques become increasingly complex, employing increasingly sophisticated and psychologically manipulative tactics. The financial loss of this crisis is staggering; according to the National Cyber Security Association (NCA) of Vietnam, online fraud caused an estimated loss of 18,900 billion VND (approximately $745 million USD) in 2024 alone [1]. A notable and alarming trend currently is the operation of large-scale scam centers in unregulated, cross-border regions. And Vietnam is not the exception; there are still numerous scam companies that target Vietnamese citizens, especially those who have leaked private information or who have low awareness of privacy and fraud. In response to this situation, the development of robust machine learning models for the automatic detection, warning, and prevention of these scams has become increasingly vital and promising. However, progress in this area is impeded by significant challenges, primarily scarcity of high-quality, domain-specific, and publicly available training data, particularly for non-English languages like Vietnamese. Moreover, our review of existing Vietnamese training dataset reveals that while datasets are available for general Natural Language Processing (NLP) tasks—such as sentiment analysis (e.g., UIT-ViCTSD), fact-checking (e.g., ViWikiFC), and Automatic Speech Recognition (e.g., VIVOS)—there is a distinct lack of resources focused on the unique characteristics of conversational, fraudulent dialogues. This gap prevents researchers from developing and benchmarking specialized models that can understand the nuances, structure, and manipulative language of scam calls.

To address this critical gap, this paper introduces **FraudTeleCallGenerator**, a systematic pipeline for generating a comprehensive, multimodal dataset of fraudulent and legitimate telephone conversations in Vietnamese. Our methodology is built upon a multi-agent system where AI agents, acting as scammers and potential victims, engage in dynamic dialogues. The system is designed to be highly realistic, simulating conversations based on 15 culturally relevant fraud scenarios and 8 distinct user profiles that reflect Vietnamese demographics. The system's realism is enhanced by its foundation on 15 fraud scenarios derived not from theoretical constructs, but from prevalent, real-world cases that are continuously documented by major national news outlets and government agencies, confirming their urgency and relevance. The resulting text-based dialogues are subsequently synthesized into a parallel audio corpus using a state-of-the-art Vietnamese Text-to-Speech (TTS) model, creating a rich resource for both text and speech-based analysis.

The main contributions of this research can be summarized as follows:

- A novel, systematic methodology for generating domain-specific, conversational data for a low-resource language. Our proposed pipeline, *FraudTeleCallGenerator*, utilizes a configurable multi-agent system to simulate realistic, goal-oriented dialogues, including a robust strategy for handling API interactions.
- A comprehensive, multi-modal dataset, to the best of our knowledge, of fraudulent telephone call dialogues for the Vietnamese language. The dataset contains parallel text and audio recordings, enriched with detailed metadata, and is grounded in 15 culturally-specific scam scenarios, making it a highly relevant resource for the research community.
- A thorough statistical analysis of the generated dataset to validate its quality and utility. This analysis confirms the diversity of the dataset's content across various scenarios, user profiles, and conversational structures, establishing a benchmark for future research in this area.

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The main contributions of this research are threefold. First, we introduce a novel, systematic methodology for generating domain-specific, conversational data for a low-resource language. Our proposed pipeline, *FraudTeleCallGenerator*, utilizes a configurable multi-agent system to simulate realistic, goal-oriented dialogues, including a robust strategy for handling API interactions. Second, we present the first comprehensive, multi-modal dataset, to the best of our knowledge, of fraudulent telephone call dialogues for the Vietnamese language. This dataset contains parallel text and audio recordings, is enriched with detailed metadata, and is grounded in 15 culturally-specific scam scenarios, making it a highly relevant resource for the research community. Finally, we provide a thorough statistical analysis of the generated dataset to validate its quality and utility, confirming its diversity across various scenarios, user profiles, and conversational structures, thereby establishing a benchmark for future research in this area.

2 Literature Review

3 Related Work

This section reviews existing work in two key areas relevant to our research: Vietnamese language processing datasets and datasets for fraud detection. We aim to contextualize our contribution and highlight the existing research gap that our work addresses.

3.1 Datasets for Language Processing in Vietnamese

The field of natural language processing (NLP) for Vietnamese has seen significant growth, leading to the development of several valuable public datasets. For speech processing, VIVOS stands out as a prominent free Vietnamese speech corpus developed for Automatic Speech Recognition (ASR) research [2]. However, ASR corpora like VIVOS typically consist of read speech, where individuals read from prepared scripts, and thus lack the spontaneous, interactive, and turn-taking nature of real telephone conversations.

Datasets such as *UIT-ViCTSD*, a prominent corpus for toxic speech detection sourced from social media comments, exemplify sentence-level classification. In this task, a model is trained to analyze the linguistic features within a single, isolated comment to assign a label (e.g., "toxic" or "constructive") [**?**]. The classification is largely independent of the broader conversational thread; the model does not need to know who wrote the previous comment or what the original topic of the post was to make its decision.

Similarly, *ViWikiFC*, a dataset designed for fact-checking, operates on a static pair of texts: a claim and a piece of evidence. The model's task is to assess the semantic relationship between these two fixed inputs to determine a veracity label (e.g., "Supported", "Refuted") [4]. Although complex, this is fundamentally a task of textual entailment or reasoning, not an analysis of a dynamic, evolving conversation.

Both of these tasks are therefore primarily **stateless**. The entire context required to make a classification decision is self-contained within the individual data sample provided to the model. They do not require the model to maintain a memory of preceding turns, track the changing intentions of participants, or understand how a dialogue strategically unfolds over time. This stands in stark contrast to the **stateful** nature of scam detection, which we will address next.

In contrast, telecommunication fraud detection is inherently a dialogue-level, stateful classification problem. A model cannot reliably determine whether a conversation is fraudulent by analyzing sentences in isolation. For instance, the utterance "Please provide the confirmation code we just sent you" is benign on its own. However, within a scam dialogue, it becomes a malicious act only when understood as the culmination of a multi-turn sequence involving tactics like impersonation (e.g., "I'm calling from the bank"), fabricating urgency (e.g., "Your account is locked"), and building false trust. Therefore, the task demands a model that can comprehend the entire goal-oriented structure, track the manipulative progression over multiple turns, and understand the contextual relationship between the scammer's prompts and the victim's responses. Our dataset is specifically constructed to capture this longitudinal, stateful nature of fraudulent conversations, a characteristic absent in existing Vietnamese NLP resources.

3.2 Datasets for Fraud and Scam Detection

The challenge of fraud detection has been extensively studied in various domains, particularly in finance, where models are trained on transactional data to detect anomalies. In the conversational domain, datasets for scam detection do exist for high-resource languages like English, yet they possess fundamental limitations for the task of modeling interactive fraud.

A prominent example is the *SMS Spam Collection* [?] from the UCI Machine Learning Repository, a widely-used benchmark dataset containing approximately 5,574 English SMS messages. While valuable for text classification, its utility for our purpose is constrained. Firstly, the spam category is heterogeneous, mixing general advertising and simple phishing with a smaller subset of fraudulent texts, thus lacking a specific focus on complex, goal-oriented financial fraud scenarios. More critically, its fundamental drawback lies in its static, non-conversational nature. Each entry is a discrete, self-contained message, which precludes the modeling of multi-turn dialogue structures, narrative progression, and the interactive, psychological manipulation that is central to a phone scam. A model trained on this data can only learn to classify a single block of text, not to understand an evolving, stateful interaction.

Similarly, for audio-based analysis, resources like the *Robocall Audio Datasets* [?] provide real-world audio samples of automated calls. The datasets are collected via the U.S. Federal Trade Commission FTC's Project Point of No Entry contains over 1,400 recordings, mostly automaton calls, where only the robocaller's voice is recorded - the recipient's speech is either omitted or not transcribed. These are valuable for developing systems that can identify the acoustic fingerprints of pre-recorded messages or synthetic voices. However, their primary limitation is their unidirectional format. These are audio recordings of a single party speaking (a monologue), not an interactive dialogue between two or more parties. Consequently, they lack the turn-taking dynamics, speaker interplay, and adaptive responses inherent in a real conversation. A model trained on robocall data could learn to classify a call as "automated," but it would be incapable of understanding the strategic, back-and-forth progression of a human-led scam, where the fraudster must constantly adapt their tactics based on the victim's real-time reactions.

The work most methodologically analogous to our own is the *TeleAntiFraud-28k* [?] dataset. The authors similarly employ a multi-agent adversarial framework with a caller, a callee, and a manager to synthetically generate dialogues. This approach has proven effective for creating a large-scale, multi-modal dataset for Mandarin Chinese. It successfully demonstrates the value of creating synthetic dialogues for this task. However, its primary and crucial limitation is its exclusive focus on Mandarin Chinese, rendering it inapplicable to the Vietnamese context. This highlights a critical point: fraud is not only language-dependent but also culturally-specific. The tactics and scenarios effective in one culture may not be in another. A simple translation of the *TeleAntiFraud-28k* dataset would fail to capture the unique linguistic expressions and, more importantly, the socio-cultural context exploited by scams prevalent in Vietnam (e.g., impersonating local authorities, specific banking procedures, etc.).

Therefore, a new resource tailored to the Vietnamese language and its unique fraud landscape is required. The 15 scenarios that form the basis of our dataset, such as impersonating police officers ("giả danh công an"), post office employees ("lừa đảo bưu điện"), or tax officials ("lừa đảo thuế"), are tailored to exploit specific societal norms, fears, and administrative procedures prevalent in Vietnam. These culturally-specific scenarios are not represented in existing English-language datasets.

3.3 Research Gap

Our review of the literature reveals a clear and critical gap: while general-purpose speech and text datasets exist for Vietnamese, there is no publicly available, multi-modal dataset specifically designed for the task of telecommunication fraud detection. Furthermore, existing fraud detection datasets in other languages lack the linguistic and cultural specificity required to be effective for the Vietnamese context. Our work aims to fill this void by providing the first, to our knowledge, comprehensive and culturally-grounded dataset for Vietnamese scam call analysis.

4 Methodology

4.1 Overall Pipeline Overview

The methodology proposed in this paper is guided by two core objectives: (i) maximizing the realism of the resulting dataset, and (ii) ensuring the highest standards of data privacy. While using real-world call recordings may appear to be a straightforward solution, such an approach introduces significant ethical and privacy concerns, particularly due to the presence of sensitive personal information, financial details, and private conversations.

To address these challenges, we introduce *FraudTeleCallGenerator*—a synthetic, multi-modal data generation pipeline designed with privacy-by-design as a foundational principle. Because the entire dataset is artificially generated, no real user data is collected or processed, thereby making the final corpus suitable for open research and public deployment.

The proposed pipeline comprises two main phases: (1) **Text-based dialogue generation** and (2) **Audio corpus synthesis**. A central emphasis throughout both phases is the procedure from generated dialogues to the complex dynamics of real-world fraudulent phone interactions. This realism is achieved through several integrated strategies: grounding dialogues in culturally specific scenarios inspired by actual scam reports, simulating conversations using a multi-agent adversarial framework that models deceptive behavior, and enhancing both linguistic and acoustic naturalness in the final outputs.

4.2 Phase 1: Text-based Dialogue Generation

4.2.1 Model conceptual foundation

4.2.1.1 Dialogue as a multi-agent system At a fundamental level, human dialogue is not merely a sequential exchange of utterances, but a complex, interactive process where participants strategically pursue specific objectives. This dynamic can be effectively modeled using the concept of a *multi-agent system* (MAS), a paradigm from artificial intelligence where multiple autonomous agents interact within a shared environment. In this paradigm, each participant in a conversation is treated as an agent possessing an internal state and a set of goals. Their actions—the words they speak—are chosen strategically to influence the state of other agents and move the conversation closer to their desired outcome.

This model is particularly well-suited for simulating fraudulent telephone calls, which are distinct from casual conversations. A scam call is an inherently goal-oriented and often adversarial interaction. The scammer-agent has a clear, deceptive objective (e.g., information extraction, financial gain), while the victim-agent's goal is typically defensive (e.g., information preservation, call termination). The conversation's state, such as the victim's level of trust or suspicion, evolves with each turn, requiring agents to adapt their strategies dynamically. Therefore, by framing the problem within a multi-agent system, we establish a robust conceptual foundation for generating dialogues that capture the strategic and stateful nature of real-world scam interactions.

4.2.1.2 Conceptual agent framework Our proposed MAS architecture consists of four distinct agents, each with a specialized role and a set of configurable attributes that govern the conversations' content and flow.

4.2.1.3 LLM-based imitation The behavioral realism of the agents in our system is achieved by leveraging the powerful imitative and role-playing capabilities inherent in modern large language models (LLMs). Rather than hard-coding conversational scripts, we employ a methodology analogous to the self-instruct paradigm by guiding the LLM's behavior through profoundly crafted system prompts that define each agent's persona, objectives, constraints, and tactical knowledge.

For the left agent, these prompts are specifically designed to mimic well-established social engineering principles commonly used in real-world fraud:

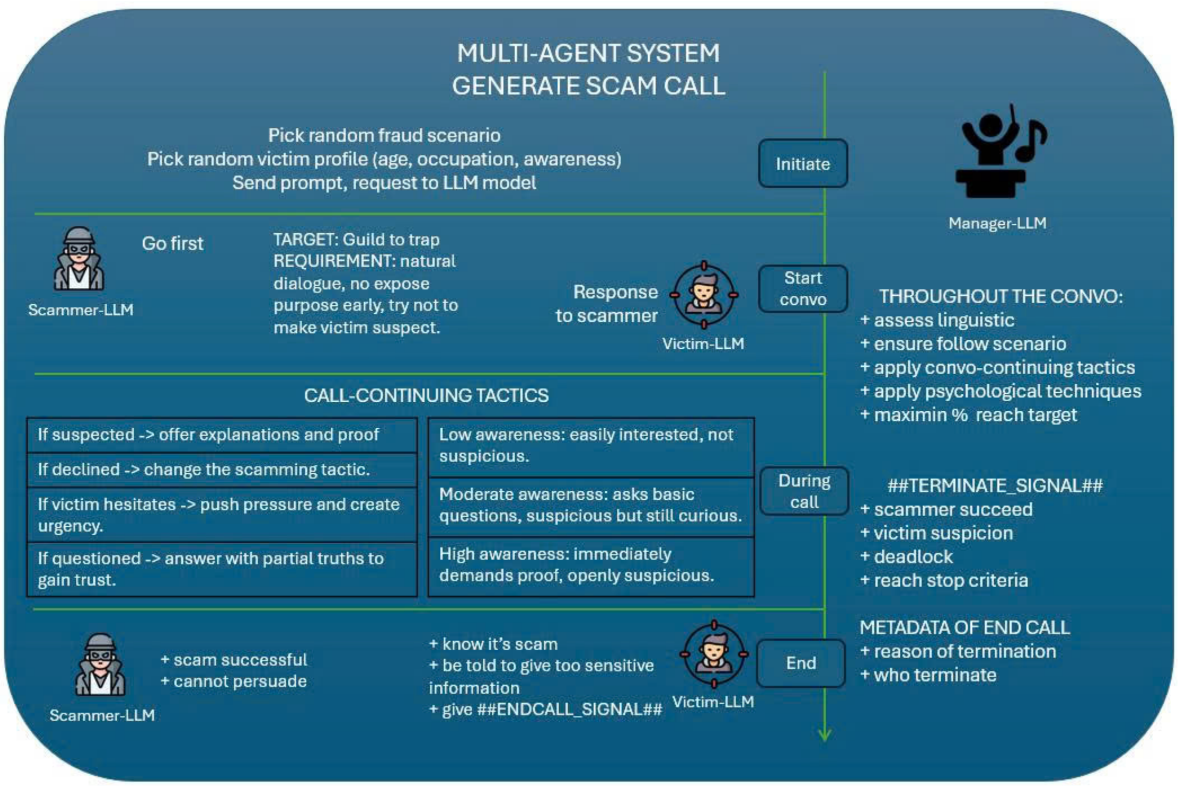
- The agent is instructed to impersonate figures of authority, such as police officers, tax officials, or bank employees, using formal and assertive language to command compliance.

- The prompts create a sense of crisis (e.g., "Your account has been compromised") or a limited-time opportunity (e.g., "This investment offer expires today"), compelling the victim to act impulsively without critical thought.
- In scenarios like romance scams, the agent is prompted to build an emotional connection, feign empathy, and establish trust before making fraudulent requests.

Conversely, the right agent's prompts are modulated by its user profile, enabling the LLM to imitate the distinct response patterns of different demographics—from the trusting nature of an elderly individual to the skepticism of a tech-savvy student. By combining this LLM-based imitation with established principles of social engineering, our framework generates dialogues that are not only linguistically coherent but also psychologically and behaviorally plausible, capturing the subtle manipulative tactics that define real scam calls.

4.2.2 Multi-agent architecture for Fraudulent call generation

Our approach employs a sophisticated multi-agent system designed to generate realistic telephone fraud conversations through carefully orchestrated interactions. We introduce a three-agent architecture supported by a central orchestration mechanism to ensure natural and diverse dialogues across multiple fraud scenarios.



Hình 1: Multi-agent system workflow for generating scam call dialogues.

4.2.2.1 System architecture and Agent specialization The core of our system comprises three specialized language model agents, each with distinct roles:

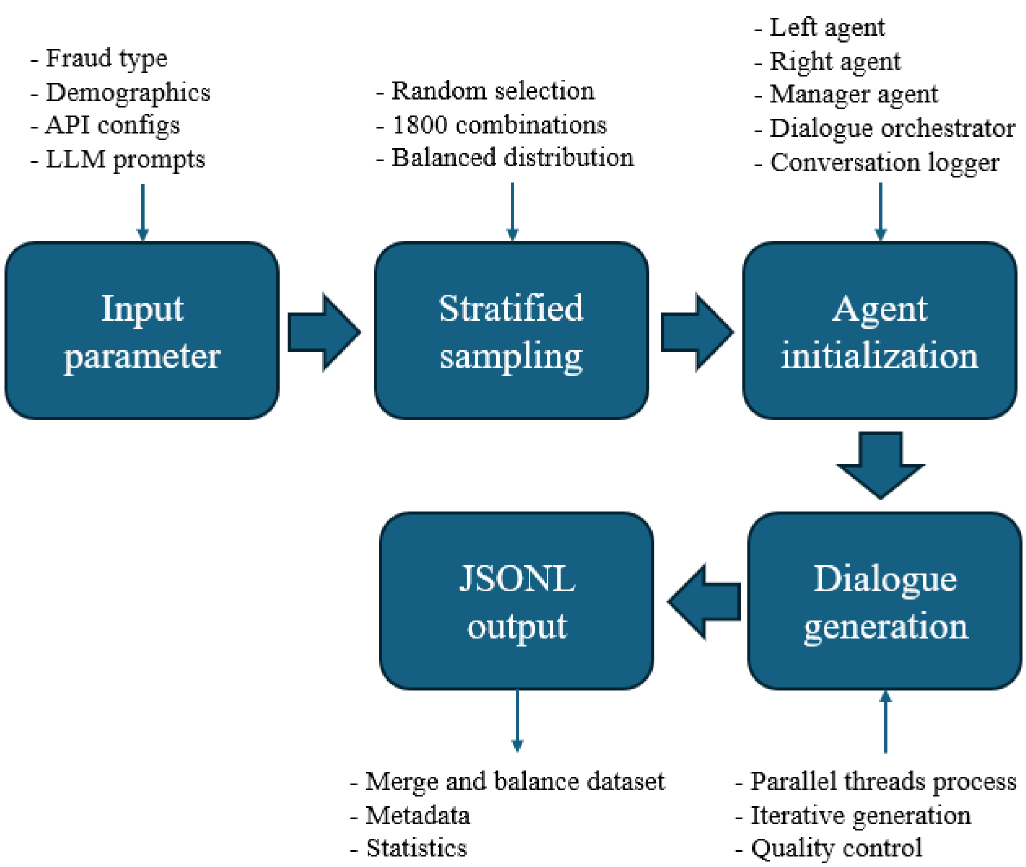
First, the *left agent*, as a scammer, plays the offensive role. Its primary objective is to proactively probe for and exploit vulnerabilities in the victim's awareness and psychological state. It actively drives the conversation toward its fraudulent goal based on a predefined attack vector.

In contrast, the *right agent*, as a victim, plays the defensive role. Its actions are reactive to the scammer, whose level of reaction or defense is determined by its configured user profile, where a higher awareness level corresponds to a more robust defensive posture.

While the paradigm of left and right agent creates dynamic simulation, unconstrained interactions between generative AI agents can lead to common pitfalls, such as repetitive conversational loops, logical incoherence, or deviation from the intended scenario. To mitigate these issues and ensure data quality, we introduce a third entity into our framework: the *manager agent*. This agent does not participate directly in the conversation but actively monitors the interaction between the two agents. Its primary responsibilities are:

- It assesses whether the dialogue remains coherent and plausible, terminating sessions that devolve into nonsensical exchanges.
- It ensures the simulation stays on track with the intended fraud scenario.
- It applies a set of termination rules to conclude the conversation when a clear outcome is reached. This includes scenarios where the scammer succeeds (the victim is deceived), the victim successfully defends (the victim expresses definitive suspicion), or the dialogue reaches a stalemate (e.g., exceeding a maximum number of turns).

4.2.2.2 Pipeline implementation



Hình 2: The workflow with step components of text dataset generator

Overview The pipeline transforms high–level configuration (fraud scenario + victim profile + API settings) into structured JSONL conversations. It is organized as five sequential stages:

Input Parameters → Stratified Sampling → Agent Initialization → Dialogue Generation → JSONL Output

Each stage is engineered for reproducibility, balance, and scalability (parallel workers + retry + quality control).

Stage 1: Input Parameters This stage ingests:

- Fraud type taxonomy (15 classes) from configuration file.
- Demographic factors: age ranges (4), awareness levels (3), occupations (10).
- LLM execution parameters: model name, base URL, API key.
- Prompt templates (scammer / victim / manager) defining role constraints and behavioral tactics.

Let $\mathcal{F}$ be the fraud type set ($|\mathcal{F}| = 15$), $\mathcal{A}$ the age range set ($|\mathcal{A}| = 4$), $\mathcal{W}$ the awareness levels ($|\mathcal{W}| = 3$), and $\mathcal{O}$ occupations ($|\mathcal{O}| = 10$). The theoretical Cartesian space of distinct victim profiles is:

$$|\mathcal{S}| = |\mathcal{F}| \times |\mathcal{A}| \times |\mathcal{W}| \times |\mathcal{O}| = 15 \times 4 \times 3 \times 10 = 1800.$$

Stage 2: Stratified Sampling Goal: approximate uniform coverage over (f, a, w) while randomly drawing an occupation o to avoid combinatorial explosion when total requested dialogues $N < 1800$.

Algorithm 1 Stratified Scenario Scheduling

Require: Requested dialogue count N

Build list

$$L = \{(f, a, w) \mid f \in \mathcal{F},\ a \in \mathcal{A},\ w \in \mathcal{W}\}$$

Let $B \leftarrow |L| = 15 \times 4 \times 3 = 180$
Base quota per triple: $q \leftarrow \lfloor N/B \rfloor$
Remainder $r \leftarrow N - q \cdot B$
Initialize empty task list T
for all $(f, a, w) \in L$ **do**
    $c \leftarrow q + \mathbf{1}[r > 0]$
    if $r > 0$ **then**
        $r \leftarrow r - 1$
    end if
    for $i = 1$ to c **do**
        Sample age $age \sim \text{Uniform}(a_{\min}, a_{\max})$
        Sample occupation $o \sim \text{Uniform}(\mathcal{O})$
        Append task (f, age, w, o) to T
    end for
end for
Shuffle T to decorrelate ordering
return T

This preserves near–balanced marginals over fraud types and awareness levels even when N is not a multiple of B.

Stage 3: Agent Initialization For each scheduled task:

- **LeftAgent** (scammer) receives fraud-type–conditioned system prompt (scenario script + adaptive tactics).
- **RightAgent** (victim) receives demographic prompt (age, awareness influencing skepticism, occupation influencing jargon).
- **ManagerAgent** receives evaluation rubric (termination criteria and classification rationale).

- **DialogueOrchestrator** binds the three agents, holds global turn loop, logging, and post–processing.

Initialization Constraints

1. All agents share the same model endpoint abstraction (OpenAI-compatible or Gemini branch).
2. Retry parameters (max attempts, backoff delay) injected to mitigate transient API failures.
3. Conversation logger attaches timestamps and preserves raw turn order for downstream auditing.

Stage 4: Dialogue Generation Loop The orchestrator executes a bounded iterative loop up to $T_{\max}$ turns (default 15–25). Each turn consists of (a) scammer utterance, (b) victim response, (c) optional manager evaluation after a warmup (e.g., from turn ≥ 2).

Algorithm 2 Dialogue Orchestration

Require: Initialized agents (L, R, M), max turns $T_{\max}$

$H \leftarrow [\,]$ ▷ history: sequence of $(role, text)$
$term \leftarrow$ False
for $t = 1$ to $T_{\max}$ **do**
  $u_L \leftarrow L.\textsc{Generate}(H)$
  Append (left, u_L) to H
  if SignalEnd(u_L) **then**
    $reason \leftarrow$ "Left issued end signal"
    $term \leftarrow$ True; $terminator \leftarrow$ left; **break**
  end if
  $u_R \leftarrow R.\textsc{Generate}(H)$
  Append (right, u_R) to H
  if SignalEnd(u_R) **then**
    $reason \leftarrow$ "Right issued end signal"
    $term \leftarrow$ True; $terminator \leftarrow$ right; **break**
  end if
  if $t \geq 3$ **then**
    $e \leftarrow M.\textsc{Evaluate}(H)$ ▷ JSON decision: should_terminate, terminator, reason
    if e.should_terminate **then**
      $term \leftarrow$ True
      $(terminator, reason) \leftarrow (e.\text{terminator}, e.\text{reason})$ **break**
    end if
  end if
end for
if $term =$ False **then**
  $reason \leftarrow$ "Reached max turns"; $terminator \leftarrow$ natural
end if
return StructuredRecord(H, terminator, reason)

Adaptive Scammer Tactics Prompts encode conditional policy:

- Suspicion → supply plausible explanation + pseudo-proof.
- Refusal → tactic shift (new angle / authority framing).
- Hesitation → urgency amplification (scarcity, temporal pressure).
- Probing questions → partial truths to increase trust.

Victim Awareness Conditioning

- Low: cooperative, minimal challenge.
- Medium: intermittent verification, still persuadable.

- High: early challenge, demands credentials, faster termination if red flags accumulate.

Termination Semantics Manager rubric checks:

1. Success (target sensitive disclosure achieved).
2. Repeated firm refusal (2–3 decisive rejections).
3. Deadlock (semantic repetition plateau).
4. Explicit end signal ("##ENDCALL_SIGNAL##").
5. Turn cap reached.

Stage 5: JSONL Output Assembly After loop exit:

- Extract parallel arrays: `left[]` and `right[]` utterances (newline normalized).
- Attach metadata: fraud_type, user_age, awareness, occupation, terminator, termination_reason, label.
- Write one JSON object per line (stream-friendly, append-safe).

Dataset Integration A higher-level generator merges fraud and normal JSONL files, shuffles, annotates class labels (`fraud`/`normal`), and computes statistics:

- Label distribution and fraud type histogram.
- Demographic marginal distributions (age bucket, occupation, awareness).
- Conversation quality metrics (average turns, average character length, turn distribution).

Resilience and quality controls

- Retry logic:
  Each agent call retries up to a max retry count with delay backoff.
- Parallelism:
  Thread pool processes independent tasks (I/O bound LLM calls).
- Logging:
  Per–session UTF-8 logs; real-time streaming of subprocess outputs; timestamps for reproducibility.
- Prompt grounding:
  Role-specific system prompts injected every generation call to reduce drift.
- Validation:
  Conversations failing structural or termination integrity can be filtered before merge.

Complexity considerations Let N be number of dialogues and T average turns. With external LLM latency L, total wall-clock under W workers approximates:

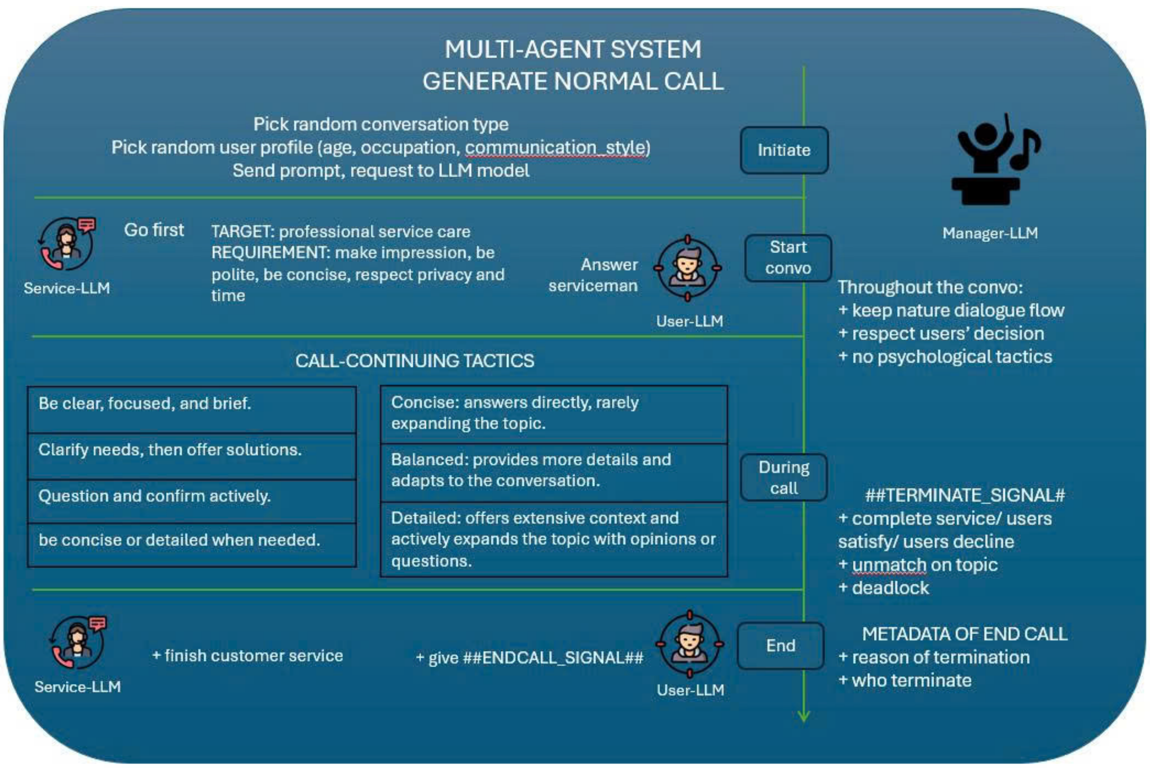
$$\text{Time} \approx \frac{N \cdot T \cdot 2L}{W} + \epsilon_{\text{overhead}}$$

(two agent calls per turn; manager every later turn adds subdominant overhead).

Pseudo-Code summary

```
for task in StratifiedSchedule(N):
    init agents (Left, Right, Manager)
    H = []
    for turn in 1..T_max:
        uL = Left.generate(H); H+=uL; if end_signal(uL): break
        uR = Right.generate(H); H+=uR; if end_signal(uR): break
        if turn>=EvalWarmup:
            dec = Manager.evaluate(H)
            if dec.should_terminate: break
    record = build_json(H, metadata, termination_context)
    write_line(record)
shuffle_and_merge_if_requested()
compute_statistics()
```

4.2.3 Multi-agent framework for Legitimate call generation



Hình 3: Multi-agent system workflow for generating scam call dialogues.

4.2.3.1 Agent roles Service Representative (LeftAgent).

Initiates every conversation with an immediate self–identification (name / department / service scope) and sets a professional tone. Its objectives are: (i) elicit the user's underlying need, (ii) clarify missing slots (time, product variant, policy detail, etc.), (iii) propose accurate solutions or actionable next steps, (iv) confirm satisfaction before closure. The response style is concise unless the user's communication style or scenario type justifies elaboration. Forbidden behaviors: applying pressure, inventing urgency, requesting unrelated sensitive personal or financial data, or digressing from the declared service scenario. When uncertainty remains, it asks targeted clarifying questions instead of guessing. If the user signals completion, it performs a structured wrap-up: summarize resolution, offer optional further help, and politely close (optionally emitting the end-call signal token if required by the pipeline).

Customer (RightAgent).
Embodies a user profile defined by (age, occupation, communication_style). Age and occupation influence lexical register (formal vs. neutral phrasing, domain vocabulary), while *communication_ style* governs verbosity and initiative:

- Concise: minimal, directly answers queries, rarely introduces new subtopics.
- Balanced: provides sufficient detail, mirrors the representative's granularity, occasionally asks follow-up questions.
- Detailed: supplies background context proactively, may branch into sub-questions, requests comparative or policy clarifications.

The customer does not fabricate resistance or suspicion heuristics (those exist only in fraud scenarios). It cooperates toward task completion, signals satisfaction explicitly, and uses the termination signal only after resolution or explicit disengagement.

ManagerAgent
Observes passively; does not inject service content. It evaluates the evolving dialogue for: (a) successful task completion (goal clearly satisfied), (b) explicit user closure (gratitude + "no further questions"), (c) deadlock (repetitive turns without semantic progress), (d) persistent off-topic drift, (e) policy breach (representative requests disallowed data), or (f) max turn budget. On a termination decision it outputs a structured JSON verdict (should_terminate, terminator, reason) and triggers a role-aligned closure prompt if needed. Any detected coercive or manipulative pattern is labeled as policy violation and forces early termination.

Dialogue Orchestrator.
Coordinates the deterministic turn order (Left $\rightarrow$ Right $\rightarrow$ optional Manager evaluation), injects system prompts per role each generation step to prevent drift, monitors for explicit end signals (`ENDCALL_SIGNAL`, `TERMINATE_SIGNAL`), and, after termination, compiles normalized utterance arrays (left[], right[]) plus metadata (conversation_type, communication_style, age, occupation, terminator, termination_reason, label=normal). It ensures one utterance per agent per turn and enforces the maximum turn cap if no earlier stopping criterion fires.

4.2.3.2 Termination semantics (legitimate setting) Valid termination reasons emitted by the manager rubric include: `service_completed`, `user_satisfied`, `deadlock`, `off_topic`, `policy_violation`, `max_turns`, or explicit user/agent signal. No fraud-oriented labels (e.g., "fraud_success", "victim_suspicion") are present in this subset.

4.2.3.3 Behavioral contrast (implicit) Distinctive signals for downstream discrimination arise from: absence of manipulative escalation, presence of structured clarification turns, cooperative lexical patterns, formal closing summary, and communication_style–driven verbosity modulation instead of awareness-driven defensive probing.

4.2.3.4 Output consistency All serialization, balancing, and statistical aggregation reuse the same JSONL schema as the fraudulent module, with field substitutions: `conversation_type` in place of `fraud_type`, and `communication_style` in place of `user_awareness`; label fixed to `normal`.

4.2.3.5 Input space differences Fraud type set $\mathcal{F}$ is replaced by a legitimate conversation type set $\mathcal{C}$ (e.g., insurance inquiry, food ordering, appointment booking, technical support). Awareness dimension $\mathcal{W}$ is replaced by communication style set $\mathcal{S}$ with $|\mathcal{S}| = 3$. Thus the Cartesian design space becomes

$$|\mathcal{S}_{\text{legit}}| = |\mathcal{C}| \times |\mathcal{A}| \times |\mathcal{S}| \times |\mathcal{O}|$$

$$
\begin{aligned}
|\mathcal{C}| &= 10 && \text{(conversation types in } \texttt{CONVERSATION\_TYPES}\text{)}\\
|\mathcal{A}| &= 4 && \text{(age buckets in } \texttt{AGE\_RANGES}\text{)}\\
|\mathcal{S}| &= 3 && \text{(communication styles reused from } \texttt{AWARENESS\_LEVELS}\text{)}\\
|\mathcal{O}| &= 10 && \text{(occupations in } \texttt{OCCUPATIONS}\text{)}\\
|\mathcal{S}_{\text{legit}}| &= 10 \times 4 \times 3 \times 10 = 1200
\end{aligned}
$$

4.2.3.6 Stratified scheduling adaptation The stratified loop identical to Algorithm 1 is applied over triples (c, a, s) instead of (f, a, w). Pseudo-replacement:

$$L_{\text{legit}} = \{(c, a, s) \mid c \in \mathcal{C}, a \in \mathcal{A}, s \in \mathcal{S}\}$$

4.2.3.7 Prompt deltas (semantic constraints) Service left prompt replaces manipulation clauses with:

1. Immediate self-identification (role + organization).
2. Clarify user need before proposing solutions.
3. No creation of urgency, no pressure escalation steps.
4. Offer resolution steps; if ambiguity remains, ask targeted clarifying questions.
5. If user declines further help, gracefully move to closure (optionally emitting end signal).

Right prompt: awareness $\to$ verbosity profile:

- Concise: minimal, direct answers, rarely expands topics.
- Balanced: adaptive detail, mirrors representative's granularity.
- Detailed: adds background context, volunteer extra constraints, may bifurcate into sub-questions.

4.2.3.8 Dialogue dynamics changes Adaptive branches that formerly reacted to suspicion / refusal / hesitation now map to service behaviors:

Unclear need $\to$ ask for specification (slot clarification).

Partial satisfaction $\to$ summarize + offer optional additional assistance.

Topic drift $\to$ gently steer back; abandon if user insists (then terminate as "off-topic closure").

Prolonged silence / repetition $\to$ attempt concise wrap-up; manager may label deadlock.

4.2.3.9 Manager termination rubric (modified conditions) Fraud "success / victim defense" states are replaced by:

1. Service goal achieved (information delivered, booking confirmed, order placed).
2. Explicit user closure (gratitude + no further need).
3. Deadlock (repetitive turns without new semantic content).
4. Irrecoverable topic drift (sustained off-scenario content).
5. Policy violation (rare; e.g., representative asks for prohibited sensitive data) — early stop.
6. Max turns cap reached.

End signal token (`ENDCALL_SIGNAL`) semantics unchanged; manager still infers terminator role.

4.2.3.10 Simplified differential pseudo-code

```
for (c,a,s) in StratifiedTriples(C, A, S):
    age  = sample_age(a); occupation = sample_occupation(age)
    Left  = ServiceAgent(c)
    Right = CustomerAgent(age, s, occupation)
    Manager = ManagerAgent(rubric="service")
    H = run_dialogue(Left, Right, Manager)
    record = serialize(H,
        conversation_type=c,
        communication_style=s,
        occupation=occupation,
        termination_reason=manager.reason)
    write_jsonl(record)
```

4.2.3.11 Fraud vs. Legitimate conversation differentiation A crucial aspect of our methodology is the differentiated approach to generating fraudulent versus legitimate conversations. While both utilize the same three-agent architecture, they differ fundamentally in agent objectives, conversation dynamics, and ethical constraints. Fraudulent conversations incorporate social engineering techniques designed to manipulate victims through false urgency, deception, and psychological pressure, while legitimate conversations emphasize professional service delivery, transparency, and respect for user autonomy. The Left Agent in fraudulent scenarios employs adaptive manipulation tactics calibrated against victim awareness levels, whereas in legitimate scenarios, it focuses on efficient, ethical service provision adjusted to user communication preferences. This intentional differentiation produces two distinct yet complementary datasets that together provide a comprehensive representation of the telecommunications dialogue landscape, enabling robust training and evaluation of fraud detection systems.

4.2.4 Optimization technique

Our implementation incorporates several engineering optimizations for scalable dataset generation:

(i) **Parallel processing:**
Utilizes ThreadPoolExecutor for concurrent dialogue generation

(ii) **Exponential backoff:**
Implements retry logic with increasing delays to handle API rate limits

(iii) **Comprehensive logging:**
Records detailed conversation histories for quality analysis

(iv) **Structured metadata:**
Generates JSONL files with rich contextual information about each dialogue

This methodology produces a diverse corpus of realistic fraud and legit conversations that captures the linguistic patterns, social engineering techniques, and psychological manipulations employed in actual telecommunications fraud scenarios.

4.3 Phase 2: Audio corpus synthesis

5 Methodology - Hieu

To generate a high-quality, diverse, and realistic conversational dataset, we developed a comprehensive social interaction simulation framework powered by a Multi-Agent System (MAS). Instead of relying on simple data generation scripts, our system orchestrates dynamic interactions between Large Language Model (LLM) powered agents, each with specific roles, goals, and behavioral patterns.

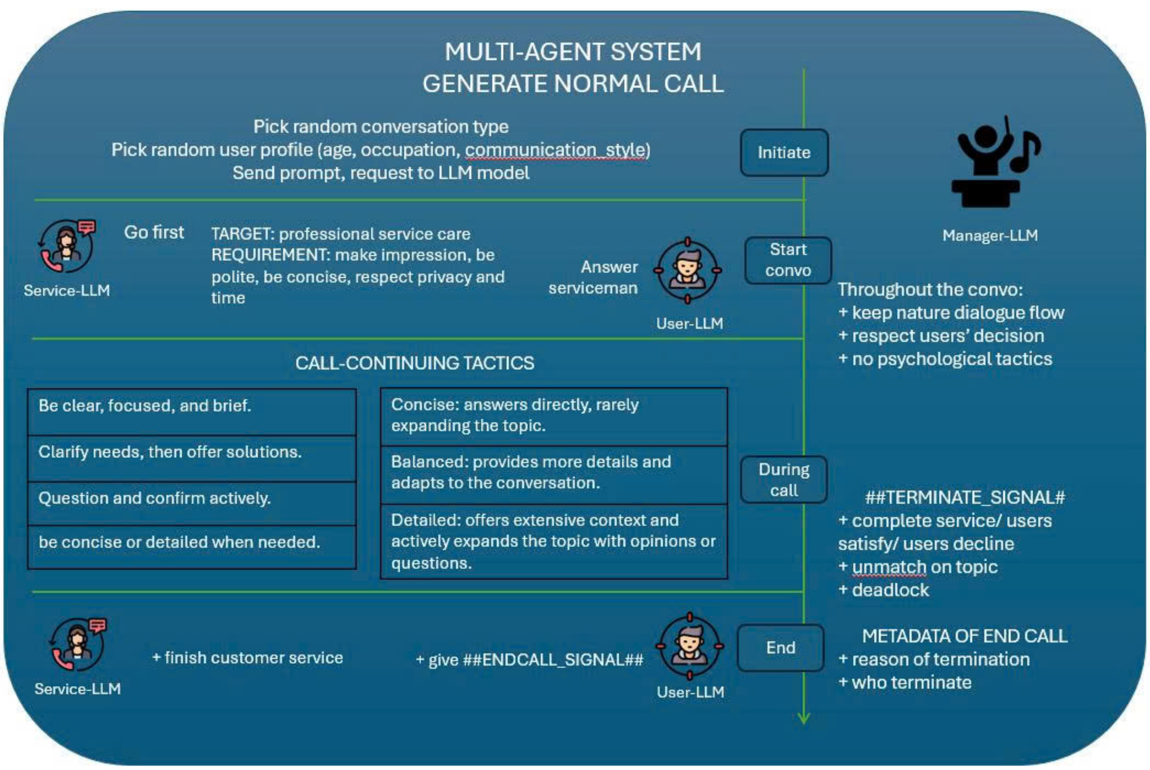
5.1 Multi-Agent System Architecture

The core of our data generation pipeline is a three-agent system designed to simulate goal-oriented telephone conversations. Each conversation involves three distinct LLM-based agents:

- **The Initiator Agent (Scammer/Service-LLM):** This agent's role is to initiate the call. In scam scenarios, it acts as the *Scammer-LLM*, with the goal of deceiving the user. In normal scenarios, it acts as the *Service-LLM*, aiming to provide professional customer service.
- **The Responder Agent (Victim/User-LLM):** This agent reacts to the initiator. As the *Victim-LLM*, it is assigned a specific persona, including demographic details (age, occupation) and a crucial "awareness level"(low, moderate, high) that dictates its susceptibility and responses to scam tactics. As the *User-LLM*, it responds based on a defined communication style (e.g., concise, balanced, detailed).
- **The Manager Agent (Manager-LLM):** This agent acts as an overseer for the entire conversation. It does not participate directly but monitors the dialogue turn-by-turn to ensure it remains coherent, on-topic, and progresses logically. It is responsible for terminating the conversation when a predefined goal is met, a deadlock occurs, or the interaction becomes unnatural.

5.2 Scam Call Generation Process

The process for generating fraudulent conversations is illustrated in Figure 4.



Hình 4: The workflow of the Multi-Agent System for generating scam calls.

The generation is initiated by randomly selecting a fraud scenario from our predefined list and a victim profile. The Scammer-LLM begins the conversation with the primary goal of guiding the victim towards the scam's objective without exposing its true purpose. Throughout the call, the Manager-LLM assesses the dialogue, applies psychological techniques to maintain realism, and works to maximize the chance of the scammer reaching their target.

The interaction dynamics are heavily influenced by the Victim-LLM's awareness level:

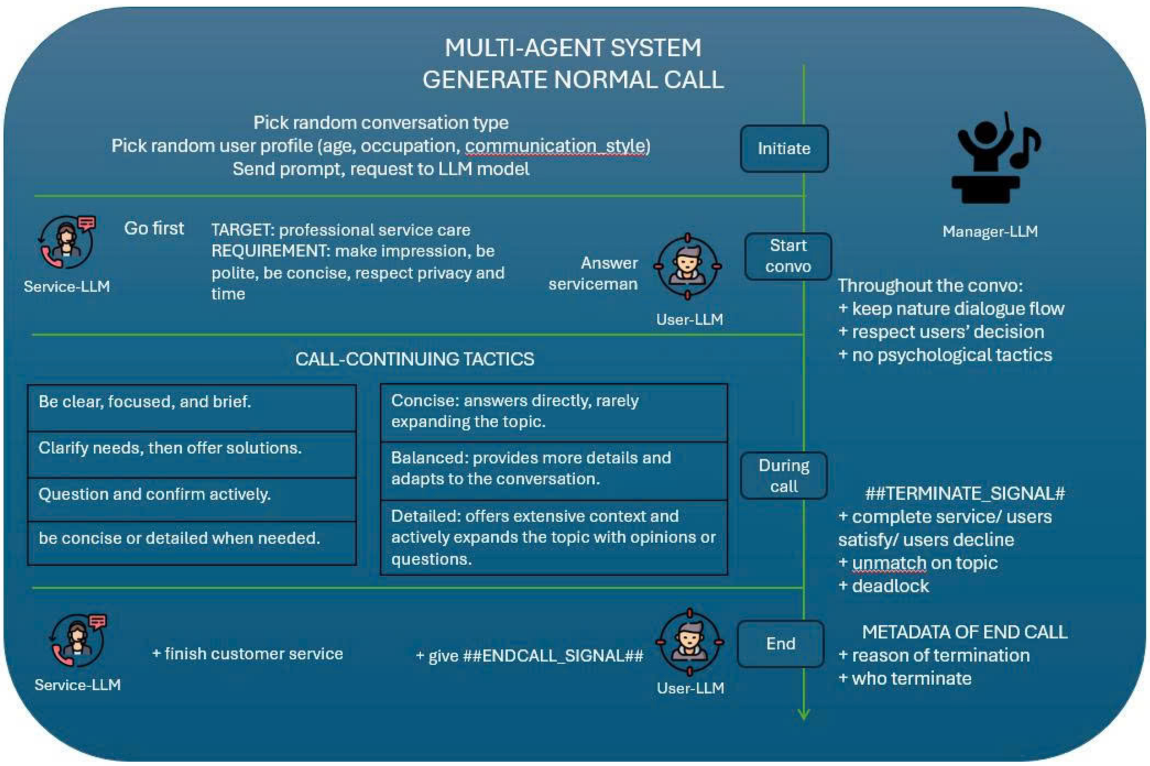
- **Low Awareness:** The victim is easily interested and not suspicious.

- **Moderate Awareness:** The victim asks basic questions and is suspicious but still curious.
- **High Awareness:** The victim immediately demands proof and is openly suspicious.

The Scammer-LLM adapts its tactics accordingly. If the victim is suspicious, it offers explanations. If the victim declines, it changes tactics. If the victim hesitates, it applies pressure and creates urgency. The conversation terminates based on signals from the Manager-LLM, such as the scammer succeeding, the victim detecting the scam, or a deadlock. Finally, metadata, including the reason for termination, is recorded.

5.3 Normal Call Generation Process

The generation of normal, non-fraudulent conversations follows a similar structure but with different objectives and tactics, as shown in Figure 5.



Hình 5: The workflow of the Multi-Agent System for generating normal calls.

Here, a conversation type (e.g., booking a service, customer support) and a user profile are selected. The Service-LLM initiates the call with the goal of providing professional, polite, and concise service. The Manager-LLM ensures the dialogue flows naturally and respects the user's decisions without applying psychological pressure. The User-LLM's communication style (concise, balanced, or detailed) dictates its response patterns. The call terminates upon successful service completion, user decline, or deadlock, with corresponding metadata being logged.

6 Experiments

Để đánh giá chất lượng và tính hữu dụng của bộ dữ liệu được tạo ra, chúng tôi đã tiến hành một loạt các phân tích định lượng chi tiết. Các phân tích này nhằm mục đích xác thực các thuộc tính của bộ dữ liệu về mặt quy mô, sự đa dạng, động lực hội thoại, và các đặc điểm ngôn ngữ riêng biệt, từ đó thiết lập một cơ sở vững chắc cho việc huấn luyện và đánh giá các mô hình phát hiện lừa đảo trong tương lai.

6.1 Dataset Statistical Analysis

6.1.0.1 Tổng quan kho dữ liệu (Corpus Overview) Kho dữ liệu cuối cùng bao gồm **1,000** hội thoại, được tạo ra một cách cân bằng tuyệt đối với **500** hội thoại lừa đảo (fraudulent) và **500** hội thoại hợp pháp (legitimate). Tổng cộng, bộ dữ liệu chứa **16,058** lượt thoại (turns), với tổng thời lượng âm thanh ước tính khoảng **350** giờ. Các thông số tổng quan được trình bày trong Bảng 1.

Bảng 1: Thống kê tổng quan về bộ dữ liệu.

Thuộc tính	**Giá trị**
Tổng số hội thoại	1,000
- Hội thoại lừa đảo	500 (50.0%)
- Hội thoại hợp pháp	500 (50.0%)
Tổng số lượt thoại	16,058
Độ dài trung bình mỗi hội thoại	16.06 lượt thoại
Tổng thời lượng âm thanh (ước tính)	≈ 350 giờ
Số lượng kịch bản lừa đảo	15
Số lượng kịch bản hợp pháp	8

6.1.0.2 Phân bổ dữ liệu (Data Distribution) Bộ dữ liệu thể hiện sự phân bổ đồng đều và đa dạng trên nhiều phương diện. 15 kịch bản lừa đảo và 8 kịch bản hợp pháp được phân bổ gần như bằng nhau, đảm bảo mô hình không bị thiên vị cho bất kỳ kịch bản cụ thể nào. Về nhân khẩu học, dữ liệu bao phủ một dải tuổi rộng từ 18 đến 70, với nhóm tuổi 18-25 chiếm tỷ lệ lớn nhất (324 mẫu). Mức độ nhận thức của nạn nhân cũng được phân bổ hợp lý giữa ba mức: thấp (356 mẫu), trung bình (334 mẫu), và cao (310 mẫu), cho phép mô phỏng nhiều phản ứng khác nhau trước các cuộc gọi lừa đảo.

6.1.0.3 Động lực hội thoại (Conversational Dynamics) Một phát hiện quan trọng từ phân tích động lực hội thoại là sự khác biệt rõ rệt về độ dài giữa hai loại cuộc gọi. Như trình bày trong Bảng 2, các cuộc gọi lừa đảo có độ dài trung bình gần gấp đôi so với các cuộc gọi hợp pháp. Điều này cho thấy kẻ lừa đảo (scammer) cần nhiều lượt thoại hơn để xây dựng kịch bản, thao túng tâm lý và thuyết phục nạn nhân, trong khi các cuộc gọi dịch vụ thông thường thường ngắn gọn và đi thẳng vào vấn đề.

Bảng 2: So sánh độ dài hội thoại giữa hai loại cuộc gọi.

Thuộc tính	**Lừa đảo**	**Hợp pháp**
Độ dài trung bình (lượt thoại)	21.28	10.84
Độ dài tối thiểu (lượt thoại)	4	4
Độ dài tối đa (lượt thoại)	51	51
Độ lệch chuẩn	8.38	(Tương tự)

Phân tích về lý do kết thúc hội thoại cũng cung cấp nhiều thông tin giá trị. Trong các kịch bản lừa đảo, có tới **38.0%** kết thúc do "Nạn nhân từ chối dứt khoát"(Persistent refusal), cho thấy sự thành công trong việc mô phỏng khả năng phòng vệ của người dùng. Tỷ lệ "Lừa đảo thành công"(Fraud success) là **7.2%**, phản ánh một tỷ lệ thực tế và cung cấp những mẫu quan trọng để mô hình học các đặc điểm của một cuộc tấn công thành công.

6.1.0.4 Phân tích Ngôn ngữ (Linguistic Analysis) Sự khác biệt giữa hai loại hội thoại còn được thể hiện rõ nét qua các đặc điểm ngôn ngữ. Trung bình, lượt thoại của kẻ lừa đảo (66.71 từ) dài hơn đáng kể so với lượt thoại của một nhân viên dịch vụ trong cuộc gọi hợp pháp (40.29 từ). Tương tự, lượt thoại của nạn nhân (41.78 từ) cũng dài hơn so với khách hàng thông thường (15.01 từ), có thể do sự bối rối, đặt câu hỏi và phản ứng lại với các tình huống phức tạp mà kẻ lừa đảo tạo ra.

Phân tích các cặp từ (bigrams) phổ biến (Bảng 3) cho thấy các cuộc gọi lừa đảo thường xoay quanh các chủ đề như "thông tin", "tài khoản", "ngân hàng", trong khi các cuộc gọi hợp pháp tập trung vào "cảm ơn", "hỗ trợ", "tư vấn". Sự khác biệt về từ vựng này là một tín hiệu mạnh mẽ, hứa hẹn tiềm năng cho các mô hình phân loại dựa trên nội dung.

Bảng 3: Các cặp từ (bigrams) phổ biến nhất trong mỗi loại hội thoại.

Lừa đảo (Top 5)	**Hợp pháp (Top 5)**
(thông tin, 3780)	(ạ dạ, 1706)
(ạ dạ, 3314)	(anh chị, 1528)
(tài khoản, 3094)	(dạ vâng, 1000)
(không ạ, 2454)	(cảm ơn, 898)
(bên em, 2006)	(không ạ, 730)

6.2 Baseline Model Evaluation

6.2.0.1 Thiết lập thí nghiệm (Experimental Setup) Nhiệm vụ của mô hình là phân loại một hội thoại hoàn chỉnh (toàn bộ các lượt thoại được nối lại) vào một trong hai nhãn: "lừa đảo"hoặc "hợp pháp". Chúng tôi chia bộ dữ liệu theo tỷ lệ 80% cho tập huấn luyện (training set) và 20% cho tập kiểm thử (test set), đảm bảo sự phân bổ đồng đều của các nhãn trong cả hai tập. Hiệu suất của các mô hình được đánh giá dựa trên các độ đo tiêu chuẩn: Accuracy, Precision, Recall, và F1-Score.

6.2.0.2 Các mô hình Baseline (Baseline Models) Chúng tôi lựa chọn hai mô hình đại diện cho hai hướng tiếp cận khác nhau trong xử lý ngôn ngữ tự nhiên:

- **TF-IDF + SVM:** Một mô hình cổ điển nhưng mạnh mẽ. Đặc trưng TF-IDF được sử dụng để chuyển đổi văn bản thành các vector số, sau đó một máy vector hỗ trợ (Support Vector Machine) được huấn luyện để thực hiện việc phân loại.
- **PhoBERT:** Một mô hình Transformer được huấn luyện trước (pre-trained) dành riêng cho tiếng Việt. Chúng tôi tinh chỉnh (fine-tune) mô hình này trên bộ dữ liệu của mình. PhoBERT có khả năng hiểu ngữ cảnh và các mối quan hệ ngữ nghĩa phức tạp trong hội thoại, phù hợp với bản chất "stateful"của bài toán.

6.2.0.3 Kết quả (Results) Kết quả thực nghiệm được trình bày trong Bảng 4. Cả hai mô hình đều cho thấy hiệu suất tốt, chứng tỏ bộ dữ liệu chứa các tín hiệu rõ ràng có thể học được. Đáng chú ý, mô hình PhoBERT vượt trội hơn hẳn, đặc biệt ở chỉ số Recall và F1-Score.

Bảng 4: Kết quả phân loại của các mô hình baseline trên tập kiểm thử.

Mô hình	**Accuracy**	**Precision**	**Recall**	**F1-Score**
TF-IDF + SVM	[]	[]	[]	[]
PhoBERT	[]	[]	[]	[]

Hiệu suất cao của mô hình PhoBERT, với F1-Score đạt []%, khẳng định rằng các mô hình học sâu có thể khai thác hiệu quả các sắc thái ngôn ngữ và cấu trúc hội thoại phức tạp có trong bộ dữ liệu của chúng tôi. Đặc biệt, chỉ số Recall cao ([]%) là cực kỳ quan trọng đối với bài toán này, vì nó thể hiện khả năng của mô hình trong việc "bắt"được phần lớn các cuộc gọi lừa đảo thực tế, qua đó giảm thiểu rủi ro cho người dùng cuối. Những kết quả này không chỉ xác thực giá trị của bộ dữ liệu mà còn thiết lập một benchmark đầu tiên đầy hứa hẹn cho các nghiên cứu tiếp theo trong lĩnh vực này.

7 Results and Discussions

8 Conclusions

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